

ENDOCRINE PHYSIOLOGY OBJECTIVES

THYROID HORMONES.

1. Know the unique hormone properties of thyroid hormones relative to other amino-acid derived hormone as well as protein and steroid hormones.
2. Know the specific cellular transport mechanisms used for iodide uptake into thyroid follicular cells from plasma and out into the colloid.
3. Be able to describe the specific steps in thyroid hormone biosynthesis from tyrosine residues including the role of thyroglobulin and specific actions of thyroidal peroxidase (TPO).
4. List and distinguish the major biosynthesized thyroid hormone products and the major thyroid gland product secreted into the blood
5. Know that the half-life of T4 is longer than T3, but that T3 is more potent than T4 and understand why.
6. Be able to explain the general routes of metabolism of T3 & T3 after secretion.
7. Explain why clinical manifestations related to impaired thyroid hormone biosynthesis are typically delayed for several weeks to months after the primary impairment develops
8. Distinguish the mechanism of action of thyroid hormones on target cells from other amino acid derived hormones and protein and steroid hormones.
9. Be able to describe the major physiological actions of thyroid hormones including the effect of thyroid hormones on metabolic rate and calorogenesis.
10. Be able to describe thyroid hormone actions on protein, carbohydrate, & lipid metabolism
11. Compare and contrast the metabolic actions of thyroid hormones with those of growth hormone, catecholamines, glucocorticoids, insulin and glucagon.
12. Be able to predict how metabolic actions are altered by hypothyroidism or hyperthyroidism.
13. Know effects of thyroid hormones on growth & development of major organ systems and be able to predict pathophysiological manifestations during hypothyroidism or hyperthyroidism.
14. Be able to describe the hypothalamic-anterior pituitary feedback mechanism that regulates thyroid hormone secretion including the basic hormone properties of TRH and TSH.
15. Be able to classify primary and secondary hyperthyroid and hypothyroid conditions based on etiology, patient symptoms, and lab findings including Graves' disease, myxedema, and neonatal (Congenital) hypothyroidism.

PARATHYROID HORMONE AND VITAMIN D: Hormonal Regulation of Ca^{++} & PO_4 .

1. Know the general distribution & functions of calcium in the body.
2. Know the compartment of body calcium directly "sensed" by calcium regulating hormones
3. Know the major hormones that regulate calcium & phosphate homeostasis and their principal target tissues.
4. Know the basic hormone properties of parathyroid hormone and compare and contrast with other endocrine hormones.

5. Know the specific actions of PTH in the kidneys including the specific sites of action along the renal tubules related to regulation of calcium and phosphate homeostasis.
6. Understand why PTH has only limited capacity to reabsorb calcium from renal tubules.
7. Rationalize the physiological benefit of PTH's action of increasing PO₄ excretion in relation to calcium homeostasis.
8. Know the specific role of PTH in the regulation of vitamin D biosynthesis.
9. Review the basic structure and histology of cortical bone as it relates to the calcium & phosphate regulatory action of hormones.
10. Know what hydroxyapatite is and its relationship with soluble bone salts and bone fluid.
11. Explain bone remodeling in terms of bone synthesis and resorption and in relation to the roles of osteoclasts, osteoblasts, and osteocytes.
12. Know the major factors that regulate PTH secretion.
13. Describe the events that occur during the rapid and slow phases of PTH actions in bone in response to acute and chronic hypocalcemia.
14. Know the pathophysiological manifestations resulting from primary hyperparathyroidism and hypoparathyroidism and relate to the physiological actions of PTH.
15. Be able to predict laboratory diagnostic parameters associated with hyperparathyroidism and hypoparathyroidism
16. Know the basic hormone properties of Vitamin D and compare and contrast with other steroid hormones as well as protein hormones.
17. Explain the biosynthesis of 1,25(OH)₂cholecalciferol from skin to the liver to kidneys.
18. Know the physiological actions of vitamin D that result in increased calcium and phosphate absorption from the intestines
19. Know the factors that regulate vitamin D biosynthesis.
20. Know the general clinical manifestations that occur as a result of rickets or osteomalacia and be able to relate these symptoms to the normal physiological actions of vitamin D
21. Know the basic hormone properties of calcitonin
22. What are the basic actions of calcitonin in relation to calcium regulation?
23. Be able to describe the dynamics and sequence of events during the integrated hormonal response (PTH, Vitamin D & calcitonin) to hypocalcemia and hypercalcemia.